Distance Metrics in KNN:

Why Euclidean Isn't Always the Answer

A Machine Learning Tutorial

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GitHub Link: **https://github.com/ANaresh1918/Individual-assignment**

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1. Introduction

One of the most intuitive machine learning algorithms is also K-Nearest Neighbours (KNN): in order to classify an unseen point, identify the K each-closest training points and vote. Yet within this simplicity lies a very important assumption - what is "closest"? This question is answered by the distance measure, and the switch of the distance measure results in different neighbours and therefore a different classification result.

This tutorial explores the behaviour of KNN as it is reshaped by three distance measures, namely Euclidean, Manhattan and Minkowski, on the Wine Classification dataset. We demonstrate that metric choice is not an insignificant implementation choice but a modelling choose with sensible effect by visualising decision boundaries, using distance contour plots, and performing comprehensive accuracy analysis.

Learning Objectives: (1) How KNN uses distance to classify. (2) Geometry Compare the geometry of various metrics. (3) Not just explain what happens to classification boundaries following choice of metrics. (4) Learn the necessity of the feature scaling in the distance-based techniques.

2. Background: The Shape of Nearness

All the three measures are members of the Minkowski family, with parameter p: d(x,y) = (Sum) |x i - y i| p) (1/p). Manhattan (p=1) adds up absolute distances that form diamondlike neighbourhoods. Euclidean (p=2) calculates straight-line distance forming circular neighbourhoods. Minkowski (p=3) forms rounded-square neighbourhoods which focus on the greatest single-dimensional difference. The parameter p determines the extent to which the metric places weight on the big differences in single features as compared to the diffuses out differences among the many features.

3. Experimental Setup

In this example, we are using the in-built Wine dataset by scikit-learn (178 samples, 13 chemical features, 3 classes). The scales of features are outrageously different alcohol scales 11-15, proline scales 278-1680 - one can see the effects of scaling in this one. Standardising all features with the use of StandardScaler, we divide the features with an 80/20 stratified split (random state=42). In the case of 2D visualisations, we consider two most discriminative features flavanoids and proline.

4. Results and Analysis

4.1 Distance Contour Plots

Unit-distance contours of both metrics are presented in figure 1. These are visualisations of the shape of a neighbourhood a region of points deemed equally close to a central point. Manhattan forms a diamond, Euclidean a circle and Minkowski p=3 a rounded square. In KNN searching a neighbour, it uses the same shapes, thus various metrics pick different neighbours.

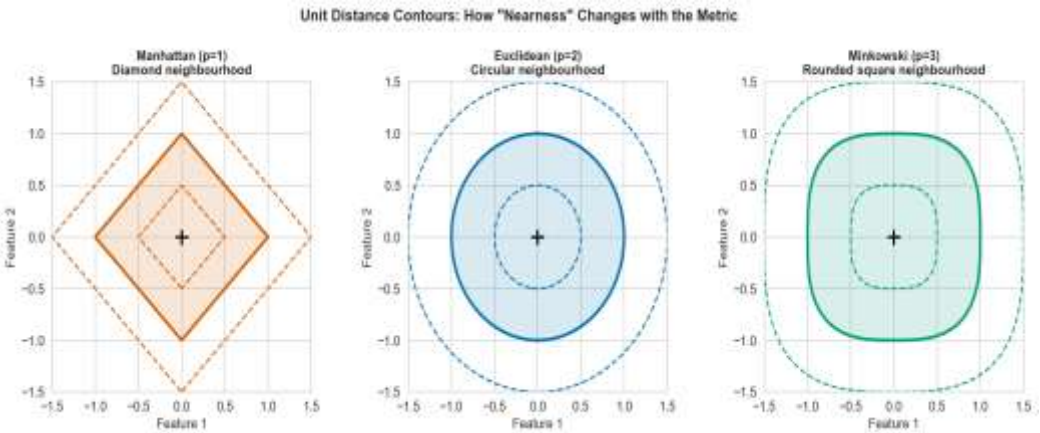

Figure 1: Unit-distance contours showing neighbourhood shapes under Manhattan (diamond), Euclidean (circle), and Minkowski p=3 (rounded square).

4.2 Decision Boundaries

Figure 2 depicts KNN decision boundaries (K=5) using the same 2D data under each of the metrics. The colour areas indicate the whereabouts of each of the classes. More importantly, the same data and K are used in these plots, with the difference in the distance metric. The borders move innocently and purposefully especially in the case of class borders where the nearest-neighbour vote is challenged. Score points close to these boundaries can be assigned various classifications based on the measure.

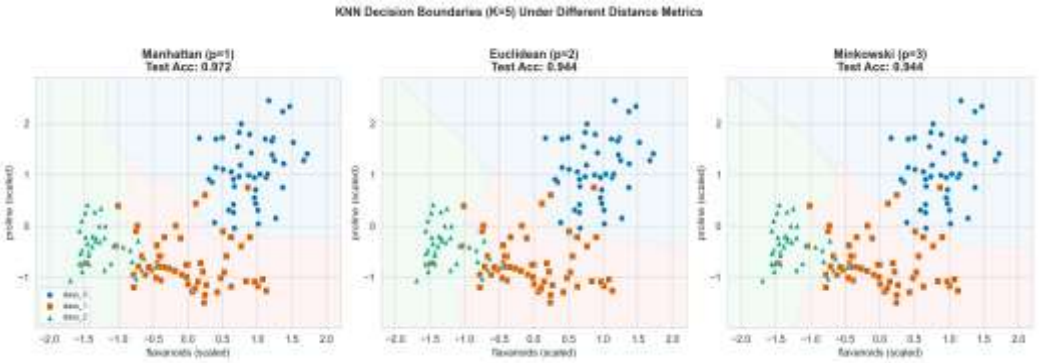

Figure 2: KNN decision boundaries (K=5) under three metrics. Same data, same K — only the metric changes.

4.3 Systematic Accuracy Comparison

Figure 3 shows a grid search on 7 values of K and 3 measures, which were assessed with 5-fold cross-validation on all 13 features. The heatmap indicates the best combinations of K×metrics. The optimum cell is brought out in the black border. It is interesting to note that the optimal metric can vary at various K values, and thus the choice of metrics and K are interacting terms that should be tuned together rather than separately.

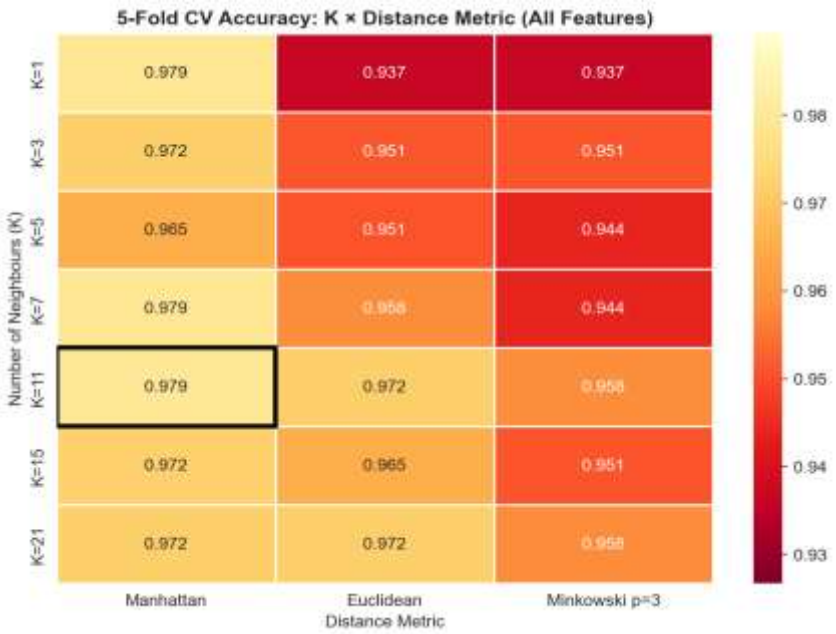

Figure 3: 5-fold CV accuracy across K values and distance metrics. Black border marks the best combination.

4.4 The Scaling Trap

Figure 4 shows the results of KNN accuracy at K=5 with and without StandardScaler preprocessing. Decline in accuracy in the absence of scaling is extreme in all measures. One characteristic of the distance is that features with high numerical values (such as proline: 2781680) dominate, and features with low numerical values are practically nonexistent (such as alcohol: 1115). Distance-based methods cannot do without feature scaling.

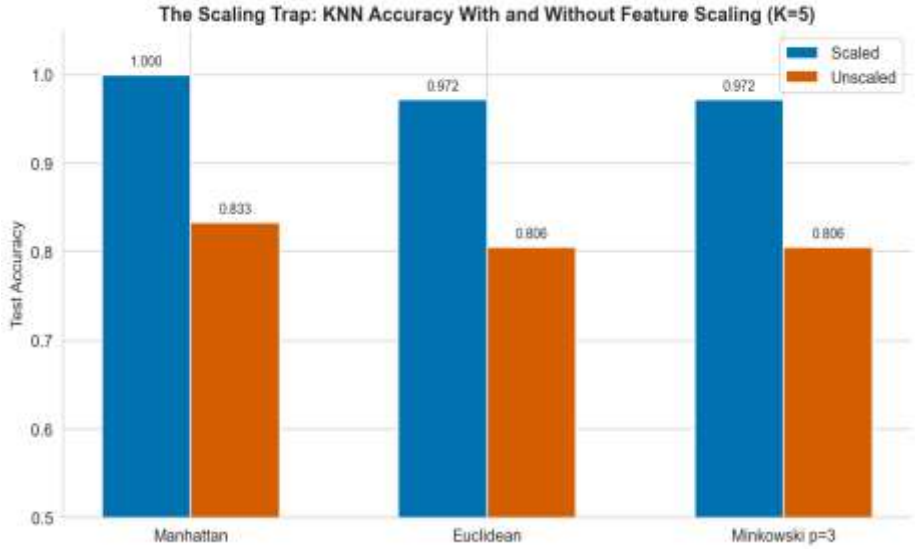

Figure 4: KNN accuracy with vs without feature scaling. All metrics suffer severely without standardisation.

4.5 Neighbour Identity Shift

Figure 5 reaches the most straight forward conclusion that metrics are important: over three test points we draw lines to their 5 closest neighbours under each metric. The coloured lines indicate that the choice of neighbours is different in different metrics based on the same point. When the various neighbours are of dissimilar classes, the classification is altered. This is particularly influential in points that are close to the boundaries of classes.

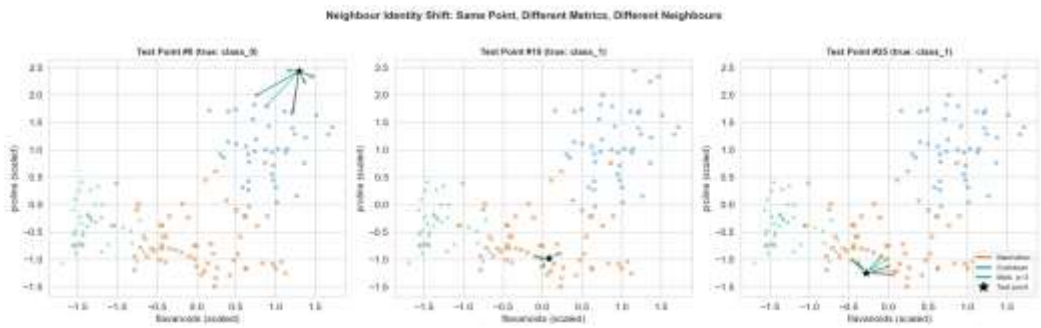

Figure 5: For three test points, lines connect to 5 nearest neighbours under each metric. Different metrics, different neighbours.

5. Choosing a Distance Metric

Scenario	Metric	Reason
General purpose	Euclidean	Default, works well with scaled features
Outliers in data	Manhattan	Less sensitive to large single-dimension differences
Many features (high-dim)	Manhattan	Distances more discriminative in high dimensions
Dominant feature axis	Minkowski (high p)	Focuses on the largest difference
Binary-like features	Manhattan	Counts total mismatches like Hamming distance

Table 1: Practical guide for choosing a KNN distance metric.

6. Limitations and Ethical Considerations

There are disadvantages to this tutorial. The boundaries of the decisions indicated only on 2 features; the entire space of 13 features does not act in the same way. Only the Minkowski family was tested, there are other distances (cosine, Mahalanobis, learned metrics). The Wine dataset consists of 178 samples only; it could be possible that larger datasets will demonstrate stronger metric impacts. We have not ventured into weighted KNN or distance based weighting.

Ethically, wine rating is not based on factuality but rather one that has his or her expert opinion. Cultural tastes are reproduced using models that are instilled using such data. When it comes to consumer facing applications, users are supposed to know that quality predictions are based on historical panels of experts. Automated quality grading may also work against small producers whose styles do not match with the norms of training data.

7. Reproducibility Notes

Random-state=42, stratified splits, StandardScaler, and built-in Wine data of scikit-learn are used in all experiments. The entire code can be seen in the notebook accompanying Jupyter. Install requirements.txt packages To reproduce: run the notebook top-to-bottom.

8. References

[1] Cover, T. & Hart, P. (1967). Nearest neighbor pattern classification. IEEE Trans. Information Theory, 13(1), 21-27.

[2] Aggarwal, C. C. et al. (2001). On the surprising behavior of distance metrics in high dimensional space. ICDT 2001.

[3] Hastie, T. et al. (2009). The Elements of Statistical Learning. 2nd Ed. Springer. Chapter 13.

[4] Scikit-learn documentation: KNeighborsClassifier. https://scikit-learn.org/

[5] Aeberhard, S. et al. (1992). Wine dataset. UCI Machine Learning Repository.